

Measuring locality boundaries in protein language model representations

A local encoder framework to quantify irreducible global dependencies

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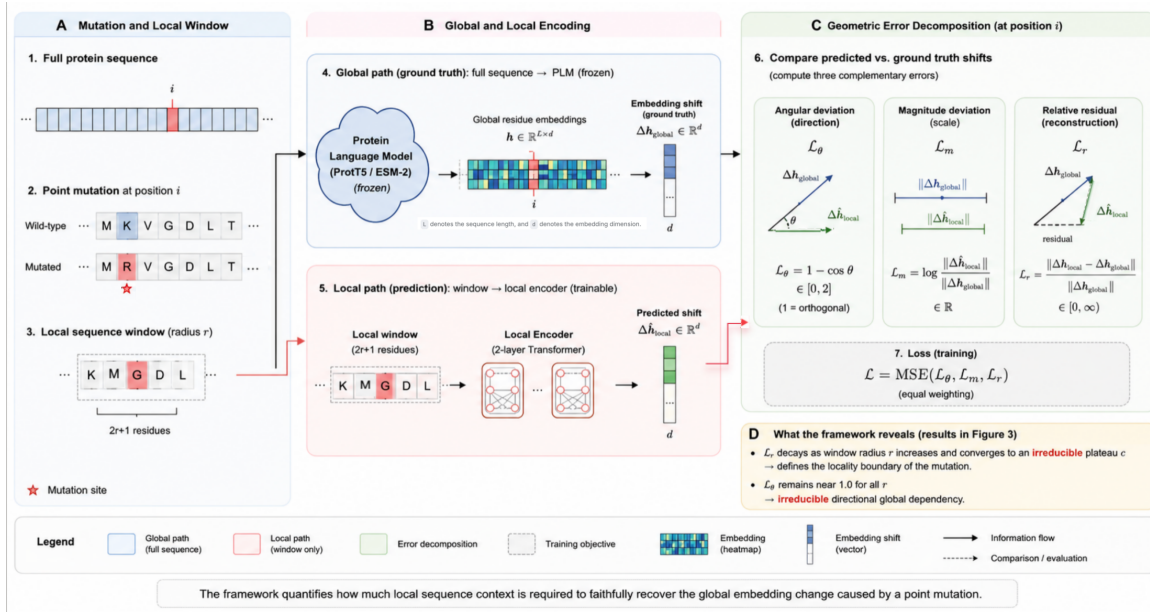


Figure 1. Local sequence window $\rightarrow$ trainable local encoder predicts Δh_{local} ; compared to Δh_{global} from full-sequence PLM (ProtT5/ESM-2). Geometric decomposition yields angular deviation L_{θ} , magnitude deviation L_m , and relative residual L_r .

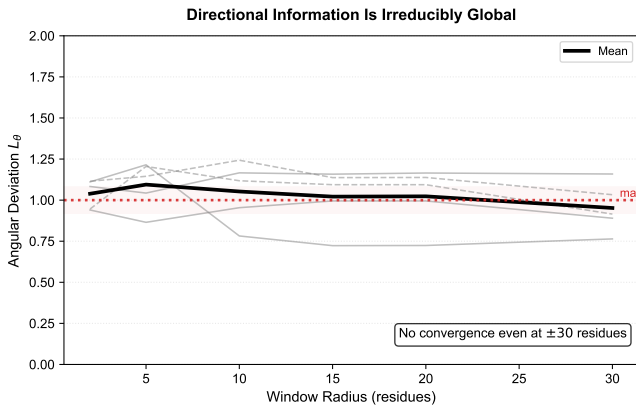


Figure 2. Directional component L_{θ} vs. window radius for five p53 hotspot mutations. L_{θ} remains near 1.0 for all radii, indicating that angular information is irrecoverable from local context.

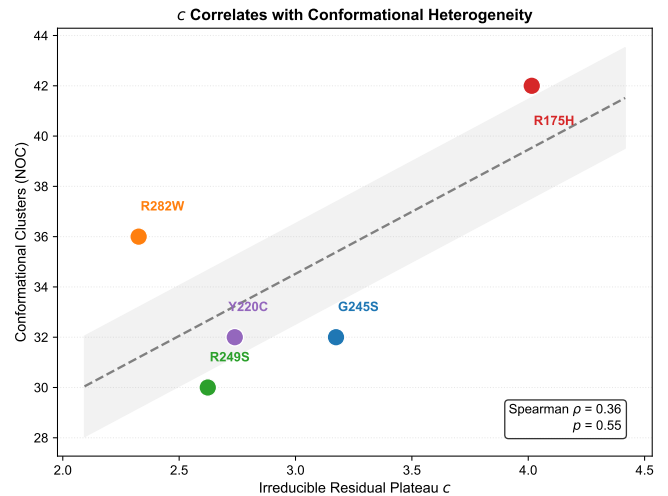


Figure 4. Plateau c vs. conformational heterogeneity (NOC clusters). Spearman $\rho \approx 0.36$, $\rho = 0.55$ (hypothesis-generating). c is orthogonal to $\Delta\Delta G$ ($\rho = -0.09$).

Key findings:

- Directional component of mutation-induced representation shifts is globally irrecoverable ($L_{\theta} \approx 1$ for all windows).
- Residual plateau c shows a positive trend with conformational heterogeneity (NOC clusters) and is orthogonal to thermodynamic stability ($\Delta\Delta G$).
- Convergence geometry is architecture-dependent: ProtT5 (monotonic decay) vs. ESM-2 (U-shaped rebound).